A proof of Seymour's second-neighborhood conjecture

Rewritten for readability from the OpenAI manuscript of the same title

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Abstract

We prove that every nonempty finite oriented graph has a vertex with at least as many vertices at directed distance two as at directed distance one. This resolves Seymour's second neighborhood conjecture positively.

1 Introduction: the result and the plan

1.1 The main result

An *oriented graph* is a finite directed graph with no loops, no multiple arcs, and no pair of opposite arcs. Its underlying undirected graph need not be complete or connected. We write $v \to u$ when there is an arc from v to u. For a vertex v, write

$$N_1^+(v) = \{u : v \to u\}, \qquad N_2^+(v) = \{w \notin \{v\} \cup N_1^+(v) : \text{ there is a vertex } u \text{ with } v \to u \to w\}.$$

Thus $N_2^+(v)$ consists of the vertices at directed distance exactly two from v. Each vertex is counted once, regardless of the number of paths reaching it.

Theorem 1.1. *Every nonempty finite oriented graph has a vertex v such that*

$$|N_1^+(v)| \le |N_2^+(v)|.$$

This is Seymour's second neighborhood conjecture. There is no connectivity or degree hypothesis. A *sink* (a vertex with $N_1^+(v) = \varnothing$) satisfies the conclusion trivially, since then $N_2^+(v) = \varnothing$ as well and $0 \le 0$. Removing from $N_2^+(v)$ the vertices already reached in one step is essential to the statement.

Sections 5 and 6 derive weighted forms and consequences for short directed cycles. Section 7 records prior work.

1.2 Image operators

For a digraph D and a set S of its vertices, define

$$F_D(S) = \{y : \text{ there is } x \in S \text{ with } x \to y\}, \qquad F_D^2(S) = F_D(F_D(S)).$$

The operator F_D does *not* subtract S: $F_D(S)$ may meet S. Clearly $F_D(S \cup S') = F_D(S) \cup F_D(S')$ and $F_D(\varnothing) = \varnothing$. In an oriented graph,

$$N_1^+(v) = F_D(\{v\}), \qquad N_2^+(v) = F_D^2(\{v\}) \setminus F_D(\{v\}). \tag{1}$$

Proof of the second identity. By definition $w \in F_D^2(\{v\})$ if and only if $v \to u \to w$ for some u. Such a w is never v, because $v \to u$ and $u \to v$ would be opposite arcs. So removing $\{v\} \cup N_1^+(v)$ from $F_D^2(\{v\})$ is the same as removing $N_1^+(v) = F_D(\{v\})$. □

1.3 Proof at a glance

Call a nonempty finite oriented graph a *counterexample* if $|N_1^+(v)| > |N_2^+(v)|$ for *every* vertex v. Theorem 1.1 says that there is no counterexample. The proof is by contradiction and has two halves, joined by a single inequality.

Half 1 (Section 2): from a counterexample to a "strictly concave" graph. If a counterexample exists, then there is an oriented graph G, all of whose vertices have an in-neighbour, such that

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad \text{for every } \varnothing \neq U \subsetneq V(G). \tag{2}$$

Equivalently $|F_G(U)| - |U| > |F_G^2(U)| - |F_G(U)|$: the first step out of U gains strictly more than the second. This is Proposition 2.1; it has two ingredients.

- *Lemma 2.2 (subset deficit).* Take a counterexample D with the fewest vertices and, among those, the fewest arcs. Then every nonempty proper vertex set S satisfies $|F_D^2(S) \setminus F_D(S)| < |F_D(S) \setminus S|$. Mechanism: deleting some arcs leaving S gives a graph that is no longer a counterexample; the vertex witnessing this must have lost out-neighbours, and bookkeeping of its lost and gained second out-neighbours gives a strict inequality for one increment. Summing the increments gives the claim.

- *Lemma 2.3 (blow-up).* Replace every vertex of D by a transitive tournament on $m > |V(D)|$ vertices, obtaining G. For $U \subseteq V(G)$ let S be the set of vertices of D whose block meets U. If $S \neq V(D)$, a deficit of at least 1 for S in D becomes a deficit of at least m for U in G, while the internal structure of the blocks costs at most $|V(D)| < m$. If $S = V(D)$ the deficit hypothesis is not available, and (2) follows directly from positive indegrees.

Half 2 (Sections 3 and 4): no such graph exists. This is Proposition 4.1. Put $X = V(G)$ and work with sets of ordered pairs in $X \times X$. A *coordinate image* advances one coordinate along an arc and keeps the other fixed: F_1 advances the first coordinate, F_2 the second. Two families $P, Q \subseteq X \times X$ are *compatible* if $F_1 P \cap F_2 Q = \varnothing$. A point in neither image is *uncovered*; let d be the number of uncovered points.

(a) Among compatible pairs with both families containing the diagonal, choose one maximizing $|P| + |Q| + d$. (The diagonal pair itself is compatible; this is the only place in Half 2 where orientedness is used. In particular the pruning lemma in (c) holds for arbitrary relations.)

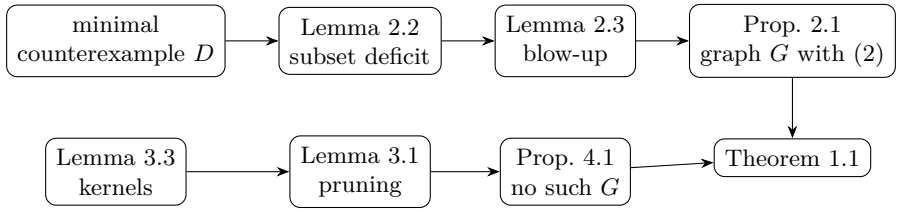
(b) Each column of P and each row of Q is a nonempty proper subset of X, so (2) applies to it. Summing, and passing to the complements $\mathcal{R} = (X \times X) \setminus F_2^2 Q$ and $\mathcal{C} = (X \times X) \setminus F_1^2 P$, gives $|\mathcal{R}| + |\mathcal{C}| > |P| + |Q| + 2d$.

(c) The new families $\mathcal{R}, \mathcal{C}$ need not be compatible. A member $(p, j) \in \mathcal{R}$ and a member $(i, s) \in \mathcal{C}$ *conflict* when $p \to i$ and $s \to j$; each conflict produces two further pairs, (p, s) and (i, j). Let H be the set of all pairs (p, s) and Z the set of all pairs (i, j) obtained in this way. The *pruning lemma* (Lemma 3.1), valid for an arbitrary finite binary relation, deletes members until no conflict remains, keeps all conflict-free members (in particular the diagonal), and deletes at most $|H| + q$ members, where q is the number of points of Z covered by no member that survives the pruning.

(d) In our situation $|H| \le d$ and $q \le d'$, where d' is the number of uncovered points of the pruned pair. Hence the pruned pair has objective value greater than $|P| + |Q| + 2d - (d + d') + d' = |P| + |Q| + d$, contradicting (a). The term d' in the deletion cost is cancelled exactly by the term d' in the new objective; this is why the objective contains the number of uncovered points and why the pruning lemma charges part of its cost to them.

Proof of the pruning lemma (Section 3). Build an auxiliary bipartite graph Γ whose vertices are the members of the two families together with two copies of the set Z. A vertex cover of Γ of size at most $|Z| + |H|$ gives the required pruning (an elementary counting argument). By Kőnig's theorem it suffices to bound a maximum matching of Γ by $|Z| + |H|$. This is done by linear algebra: four matrices L, N, B, A with $NL = AB$ factor one matrix through $\mathbb{R}^Z$ and through $\mathbb{R}^H$; for a suitably generic choice of real entries, $\operatorname{rank} L$ and $\operatorname{rank} N$ equal the sizes of two parts of the matching, and the third part gives independent vectors in $\ker A$. The last point uses an elementary rank lemma (Lemma 3.3).

Dependencies.



Sections 2 and 3 are independent of each other. Section 4 uses only the statements of Proposition 2.1 and Lemma 3.1.

1.4 Conventions and renamed symbols

$\sqcup$ denotes a disjoint union and $\subsetneq$ a proper subset. Vectors are bold; for a finite set $\mathcal{I}$, $\mathbb{R}^{\mathcal{I}}$ is the space of real vectors indexed by $\mathcal{I}$, with standard basis $(\mathbf{e}_x)$ and coordinates u_x of $\mathbf{u}$. A matrix $\mathbb{R}^{\mathcal{J}} \to \mathbb{R}^{\mathcal{I}}$ has rows indexed by $\mathcal{I}$ and columns by $\mathcal{J}$.

The source manuscript reuses several letters. To give each symbol one meaning they are renamed as follows; the mathematics is unchanged. In Lemma 2.2: Q, C, d_1, d_2 become Y, Ψ, ν_1, ν_2. In Lemma 2.3: T_v, H_0 become U_v, S^*. In Section 3: the matrices $D, D^{p,i}$, supports $T, T^{p,i}$ and index sets I, J, S_i become $W, W^{p,i}, \Theta, \Theta^{p,i}, \mathcal{I}, \mathcal{J}, \Sigma_i$; the matchings α, β and their endpoint sets R_α, C_β become $\mathcal{M}_{\mathcal{R}}, \mathcal{M}_{\mathcal{C}}, \mathcal{R}_1, \mathcal{C}_1$; the reached sets W_L, W_R become $\mathcal{V}_L^\star, \mathcal{V}_R^\star$. In Section 4 the abbreviation $M = |P| + |Q|$ is written out. In the corollaries, $A(D), M_D, d^{\pm}, Z(v), r$ become $\operatorname{Arc}(D), \Pi_D, \deg^{\pm}, \operatorname{Non}(v), \varrho$.

2 From a counterexample to a strict subset inequality

Throughout this section $F = F_D$ is the image operator of the base graph D under discussion, and F_G that of the blown-up graph G.

Proposition 2.1 (Counterexample reduction)**.** *If Theorem 1.1 has a counterexample, then there is a finite nonempty oriented graph G in which every vertex has positive indegree and*

$$|U| + |F_G^2(U)| < 2|F_G(U)| \qquad \text{for every } \varnothing \neq U \subsetneq V(G). \tag{3}$$

The proof, at the end of the section, combines Lemmas 2.2 and 2.3.

Context (not used in the proofs). The arc-deletion argument below adapts the proof idea of [20, Lemma 4]. Specializing Seacrest's parameter λ to 1, his statement gives $|F^2(S)\setminus F(S)| < |F(S)|$ in an edge-minimal counterexample when $F(S)$ is nonempty. Lemma 2.2 replaces this right-hand bound by the smaller quantity $|F(S) \setminus S|$ for nonempty proper S, under the stronger choice of minimum order and then minimum arc count. The comparison is for orientation only: Seacrest's lemma is not invoked, and Lemma 2.2 is proved in full here.

Lemma 2.2 (Subset deficit)**.** *Suppose that counterexamples exist, that is, nonempty finite oriented graphs with $|N_1^+(v)| > |N_2^+(v)|$ for every vertex v. Let D be a counterexample with the minimum number of vertices and, among the counterexamples with that number of vertices, the minimum number of arcs. Then D is strongly connected, every vertex of D has positive indegree, and*

$$|F^2(S) \setminus F(S)| < |F(S) \setminus S| \qquad \text{for every } \varnothing \neq S \subsetneq V(D). \tag{4}$$

Where it is used: as the hypothesis of Lemma 2.3.

Plan of the proof. After three easy consequences of minimality, we fix S, split $F(S)$ into the part I inside S and the part E outside, and introduce a function g on subsets of E with $g(\varnothing) = 0$ and $g(E) = |F^2(S) \setminus F(S)|$. The core is an *increment claim*: any $T \subsetneq E$ can be enlarged by a nonempty set $Y \subseteq E \setminus T$ with $g(T \cup Y) - g(T) < |Y|$. Growing T from $\varnothing$ to E and adding up gives $g(E) < |E|$.

Proof. **Step 1: D has no sink and at least two vertices.** A sink v has $|N_1^+(v)| = |N_2^+(v)| = 0$, so it violates the strict inequality $|N_1^+(v)| > |N_2^+(v)|$ that every vertex of a counterexample satisfies. Hence D has no sink; since there are no loops, D has at least two vertices.

Step 2: D is strongly connected. The strong components of D, ordered by reachability, form an acyclic digraph, so there is a strong component with vertex set V_0 such that no arc leaves V_0. For $v \in V_0$ every arc from v ends in V_0, and so does every two-step path from v. Therefore $N_1^+(v)$ and $N_2^+(v)$ are the same in D and in the induced subgraph $D[V_0]$, and $D[V_0]$ is a counterexample. By minimality of the order, $V_0 = V(D)$.

Step 3: every vertex has an in-neighbour. Given v, pick a vertex $x \neq v$ (Step 1). By Step 2 there is a directed path from x to v; its last arc enters v.

Step 4: set-up for a fixed S. Fix $\varnothing \neq S \subsetneq V(D)$ and put

$$I = F(S) \cap S, \qquad E = F(S) \setminus S, \qquad g(T) = |F(T) \setminus (I \cup T)| \quad (T \subseteq E).$$

Thus $F(S) = I \sqcup E$, $I \subseteq S$ and $E \cap S = \varnothing$. Moreover $E \neq \varnothing$: take $x \in S$ and $y \notin S$; a directed path from x to y (Step 2) contains an arc from S to the complement of S, and its head lies in E.

Step 5: increment claim. *For every $T \subsetneq E$ there is a nonempty set $Y \subseteq E \setminus T$ such that*

$$g(T \cup Y) - g(T) < |Y|. \tag{5}$$

The set Y, and the vertex v used to define it, depend on T. Fix $T \subsetneq E$.

(5a) The graph D_T and the vertex v. Let D_T be obtained from the original graph D by deleting all arcs $x \to y$ with $x \in S$ and $y \in E \setminus T$. Since $E \setminus T \neq \varnothing$ and every vertex of $E \subseteq F(S)$ is the head of an arc from S, at least one arc is deleted. So D_T is a nonempty oriented graph with the same number of vertices as D and strictly fewer arcs. By the choice of D (fewest arcs among

the counterexamples with $|V(D)|$ vertices), D_T is not a counterexample. Writing $\widetilde{N}_1, \widetilde{N}_2, \widetilde{F}$ for N_1^+, N_2^+, F computed in D_T, there is therefore a vertex v with

$$|\widetilde{N}_1(v)| \le |\widetilde{N}_2(v)|. \tag{6}$$

Since every arc and every path of D_T is also one of D, we have $\widetilde{N}_1(x) \subseteq N_1^+(x)$ and $\widetilde{F}^2(\{x\}) \subseteq F^2(\{x\})$ for every vertex x.

(5b) The vertex v lies in S and loses an out-neighbour. Suppose $\widetilde{N}_1(v) = N_1^+(v)$. Then, by (1) in D_T and in D,

$$\widetilde{N}_2(v) = \widetilde{F}^2(\{v\}) \setminus \widetilde{N}_1(v) \subseteq F^2(\{v\}) \setminus N_1^+(v) = N_2^+(v),$$

so $|N_1^+(v)| = |\widetilde{N}_1(v)| \le |\widetilde{N}_2(v)| \le |N_2^+(v)|$, contradicting that D is a counterexample. Hence some arc leaving v was deleted. Deleted arcs start in S and end in $E \setminus T$, so $v \in S$ and

$$Y := N_1^+(v) \cap (E \setminus T) \neq \varnothing, \qquad \widetilde{N}_1(v) = N_1^+(v) \setminus Y.$$

(5c) Location of the out-neighbours. As $v \in S$, $N_1^+(v) \subseteq F(S) = I \cup T \cup (E \setminus T)$, hence

$$N_1^+(v) \subseteq I \cup T \cup Y, \qquad \widetilde{N}_1(v) \subseteq I \cup T, \qquad Y \cap (I \cup T) = \varnothing.$$

The last relation holds because $Y \subseteq E \setminus T$ and $I \cap E = \varnothing$.

(5d) Endpoints of two-step paths in D_T. We claim

$$\widetilde{F}^2(\{v\}) \subseteq I \cup T \cup F(T). \tag{7}$$

Let $v \to u \to w$ be a path in D_T. By (5c), $u \in I \cup T$. If $u \in I \subseteq S$, then in D all arcs from u end in $F(S) = I \cup T \cup (E \setminus T)$, and those ending in $E \setminus T$ were deleted; so $w \in I \cup T$. If $u \in T$, then $w \in F(T)$.

(5e) Gained second out-neighbours: $\widetilde{N}_2(v) \setminus N_2^+(v) \subseteq Y \cap F(T)$. Let $w \in \widetilde{N}_2(v) \setminus N_2^+(v)$. Then $w \in \widetilde{F}^2(\{v\}) \subseteq F^2(\{v\})$. Since $w \notin N_2^+(v) = F^2(\{v\}) \setminus N_1^+(v)$, necessarily $w \in N_1^+(v)$. Since $w \in \widetilde{N}_2(v)$, we have $w \notin \widetilde{N}_1(v) = N_1^+(v) \setminus Y$. Hence $w \in Y$. Finally $w \in I \cup T \cup F(T)$ by (7) and $w \notin I \cup T$ by (5c), so $w \in F(T)$.

(5f) Lost second out-neighbours. Define

$$\Psi = F(Y) \setminus \big(I \cup T \cup Y \cup F(T)\big).$$

Then $\Psi \subseteq N_2^+(v) \setminus \widetilde{N}_2(v)$. Indeed, let $w \in \Psi$. There is $y \in Y \subseteq N_1^+(v)$ with $y \to w$, so $w \in F^2(\{v\})$; and $w \notin I \cup T \cup Y \supseteq N_1^+(v)$ by (5c). So $w \in N_2^+(v)$ by (1). (Orientedness enters here through (1): $w \neq v$ because an arc $y \to v$ would oppose $v \to y$.) On the other hand $w \notin I \cup T \cup F(T)$, so $w \notin \widetilde{F}^2(\{v\}) \supseteq \widetilde{N}_2(v)$ by (7).

(5g) Counting. By (5e) and (5f), a vertex of $\widetilde{N}_2(v)$ either lies in $N_2^+(v)$, and then not in Ψ, or lies in $Y \cap F(T)$:

$$\widetilde{N}_2(v) \subseteq \big(N_2^+(v) \setminus \Psi\big) \cup \big(Y \cap F(T)\big).$$

Write $\nu_1 = |N_1^+(v)|$ and $\nu_2 = |N_2^+(v)|$, so $\nu_2 < \nu_1$. Using (5b), (6), and $\Psi \subseteq N_2^+(v)$,

$$\nu_1 - |Y| = |\widetilde{N}_1(v)| \le |\widetilde{N}_2(v)| \le \nu_2 - |\Psi| + |Y \cap F(T)|. \tag{8}$$

(5h) The change in g is exactly $|\Psi| - |Y \cap F(T)|$. Put $\Omega = F(T) \setminus (I \cup T \cup Y)$. Since $Y \cap (I \cup T) = \varnothing$,

$$F(T) \setminus (I \cup T) = \Omega \sqcup \big(F(T) \cap Y\big), \qquad \text{so} \qquad g(T) = |\Omega| + |Y \cap F(T)|.$$

Since $F(T \cup Y) = F(T) \cup F(Y)$, and the part of $F(Y) \setminus (I \cup T \cup Y)$ outside $F(T)$ is by definition Ψ,

$$F(T \cup Y) \setminus (I \cup T \cup Y) = \Omega \sqcup \Psi, \qquad \text{so} \qquad g(T \cup Y) = |\Omega| + |\Psi|.$$

(5i) Conclusion of the claim. By (5h), then (8) rearranged, then $\nu_2 < \nu_1$,

$$g(T \cup Y) - g(T) = |\Psi| - |Y \cap F(T)| \leq |Y| + \nu_2 - \nu_1 < |Y|.$$

Step 6: summing the increments. Set $T_0 = \varnothing$. As long as $T_k \neq E$, apply Step 5 to $T = T_k$ (each time deleting arcs from the *original* graph D, so that Step 5 applies verbatim), obtain a nonempty $Y_k \subseteq E \setminus T_k$, and set $T_{k+1} = T_k \sqcup Y_k$. The sets Y_k are nonempty and pairwise disjoint subsets of the finite set E, so the process stops at some $T_r = E$ with $r \geq 1$ (as $E \neq \varnothing$), and $\sum_{k<r} |Y_k| = |E|$. Telescoping (5),

$$g(E) - g(\varnothing) = \sum_{k<r} \big(g(T_{k+1}) - g(T_k)\big) < \sum_{k<r} |Y_k| = |E|.$$

Also $g(\varnothing) = |F(\varnothing) \setminus I| = 0$. Hence $g(E) < |E|$.

Step 7: identification of $g(E)$. Since $F(S) = I \cup E$ we have $F^2(S) = F(I) \cup F(E)$, and $F(I) \subseteq F(S)$ because $I \subseteq S$. Therefore

$$F^2(S) \setminus F(S) = \big(F(I) \cup F(E)\big) \setminus (I \cup E) = F(E) \setminus (I \cup E),$$

whose cardinality is $g(E)$. With Step 6 and $|E| = |F(S) \setminus S|$ this is (4). □

We next amplify the integer deficit in (4). Each unit of the base deficit contributes the entire block size; the remaining internal contribution costs at most one unit per base vertex. The construction is the lexicographic product of D with a transitive tournament. The same graph-product construction appears in [3, Theorem 4.3]; here we need an estimate for every nonempty proper subset, rather than only for individual vertices.

Lemma 2.3 (Transitive-tournament blowup)**.** *Let D be an oriented graph on $n \geq 1$ vertices, all of positive indegree, satisfying* (4) *for every nonempty proper subset of $V(D)$. For an integer $m > n$, replace each vertex by a transitive tournament on m vertices. For every arc $v \to w$ of D, put all arcs from the block at v to the block at w, and put no other arcs between distinct blocks. The resulting graph G is oriented, all its indegrees are positive, and it satisfies* (3)*.*

Where it is used: in the proof of Proposition 2.1.

Plan of the proof. For $U \subseteq V(G)$ let S be the set of blocks that U meets. We compute $|F_G(U)|$ exactly and bound $|F_G^2(U)|$ from above, block by block. Full blocks contribute multiples of m governed by the deficit of S in D; partial blocks contribute at most 1 each. The case $S = V(D)$ is treated separately.

Proof. As everywhere in this section, $F = F_D$ is the image operator of the base graph D (applied to sets of vertices of D), and F_G is that of G. Explicitly, $V(G) = V(D) \times \{1, \ldots, m\}$, and $(v, a) \to (w, b)$ if and only if either $v = w$ and $a < b$ (an *internal* arc), or $v \to w$ in D (an *external* arc). The *block* of v is $\{v\} \times \{1, \ldots, m\}$.

Step 1: the internal image operator. For a set $\Lambda \subseteq \{1, \ldots, m\}$ of labels let $f(\Lambda) = \{b : a < b \text{ for some } a \in \Lambda\}$. Thus $f(\varnothing) = \varnothing$ and $f(\Lambda) = \{\min \Lambda + 1, \ldots, m\}$ for $\Lambda \neq \varnothing$. For nonempty Λ we have $|\Lambda| \leq m - \min \Lambda + 1 = |f(\Lambda)| + 1$ and $f^2(\Lambda) = f(f(\Lambda)) \subseteq f(\Lambda)$, hence

$$|\Lambda| - |f(\Lambda)| \leq 1, \qquad |f^2(\Lambda)| \leq |f(\Lambda)|, \qquad |\Lambda| - 2|f(\Lambda)| + |f^2(\Lambda)| \leq 1. \tag{9}$$

Step 2: G is oriented with positive indegrees. There are no loops. Two opposite internal arcs would need $a < b$ and $b < a$; two opposite external arcs would project to opposite arcs of D; an internal and an external arc never join the same pair of vertices. Every vertex (w, b) receives an arc from $(v, 1)$, where v is an in-neighbour of w in D.

Step 3: notation for a fixed U. Fix $\varnothing \neq U \subsetneq V(G)$. For $v \in V(D)$ let $U_v = \{a : (v, a) \in U\}$, and put

$$S = \{v \in V(D) : U_v \neq \varnothing\}, \qquad S^* = F(S) \cup F^2(S).$$

Then $S \neq \varnothing$ and $|U| = \sum_{v \in S} |U_v|$.

Step 4: the first image, exactly. Let $w \in V(D)$. If $w \in F(S)$, choose $v \in S$ with $v \to w$ and $(v, a) \in U$; then (v, a) sends an arc to every vertex of the block of w, so the whole block lies in $F_G(U)$. If $w \notin F(S)$, no external arc from U enters the block of w, so $F_G(U)$ meets this block exactly in $\{w\} \times f(U_w)$; this is empty when $w \notin S$. Hence

$$|F_G(U)| = m|F(S)| + \sum_{v \in S \setminus F(S)} |f(U_v)|. \tag{10}$$

Step 5: the second image, from above. We claim

$$|F_G^2(U)| \leq m|S^*| + \sum_{v \in S \setminus S^*} |f^2(U_v)|. \tag{11}$$

Blocks of vertices in S^* contribute at most m each. Let $w \notin S^*$ and let $y \in F_G^2(U)$ lie in the block of w; choose $x \in U$ and z with $x \to z \to y$, and let $v \in S$ and u be the base vertices of x and z.

- If $x \to z$ is external, then $u \in F(S)$. If $z \to y$ is external then $w \in F(\{u\}) \subseteq F^2(S)$; if internal then $w = u \in F(S)$. Both contradict $w \notin S^*$.
- If $x \to z$ is internal, then $u = v$. If $z \to y$ is external then $w \in F(\{v\}) \subseteq F(S)$, again impossible. So both arcs are internal and $w = v \in S \setminus S^*$. Writing $x = (v, a)$, $z = (v, c)$, $y = (v, b)$, we have $a \in U_v$ and $a < c < b$; so $c \in f(U_v)$ and $b \in f(f(U_v)) = f^2(U_v)$.

So the block of $w \notin S^*$ contains at most $|f^2(U_w)|$ vertices of $F_G^2(U)$, and none if $w \notin S$. This proves (11).

Step 6: combining. From Step 3, (10) and (11),

$$\begin{aligned} |U| - 2|F_G(U)| + |F_G^2(U)| \leq m\big(|S^*| - 2|F(S)|\big) + \sum_{v \in S} |U_v| \\ - 2 \sum_{v \in S \setminus F(S)} |f(U_v)| + \sum_{v \in S \setminus S^*} |f^2(U_v)|. \end{aligned} \tag{12}$$

Since $F(S) \subseteq S^*$, we have $S \setminus S^* \subseteq S \setminus F(S)$, and enlarging the last sum to $S \setminus F(S)$ only adds nonnegative terms. Now split $\sum_{v \in S} |U_v|$ over $S \cap F(S)$ and $S \setminus F(S)$; use $|U_v| \leq m$ on $S \cap F(S)$ and the third inequality of (9) (valid because $U_v \neq \varnothing$) on $S \setminus F(S)$:

$$\begin{aligned} |U| - 2|F_G(U)| + |F_G^2(U)| &\leq m\big(|S^*| - 2|F(S)| + |S \cap F(S)|\big) + |S \setminus F(S)| \\ &\leq m\big(|F^2(S) \setminus F(S)| - |F(S) \setminus S|\big) + n. \end{aligned}$$

The last line uses $|S^*| = |F(S)| + |F^2(S) \setminus F(S)|$, $|F(S)| - |S \cap F(S)| = |F(S) \setminus S|$, and $|S \setminus F(S)| \leq n$.

Step 7: the case $S \subsetneq V(D)$. Then S is nonempty and proper, so by (4) the parenthesized difference is an integer ≤ -1. The bound of Step 6 is at most $-m + n < 0$, which is (3).

Step 8: the case $S = V(D)$. Here (4) is not available. Positive indegrees in D give $F(S) = V(D)$, so $F_G(U) = V(G)$ by Step 4. Positive indegrees in G (Step 2) give $F_G^2(U) = F_G(V(G)) = V(G)$. Hence

$$|U| - 2|F_G(U)| + |F_G^2(U)| = |U| - |V(G)| < 0,$$

because U is proper. □

Proof of Proposition 2.1. Choose a counterexample D of minimum order and then minimum number of arcs. Lemma 2.2 supplies positive indegrees and the subset deficit (4), so Lemma 2.3, with any integer $m > |V(D)|$, supplies the required graph G. □

3 Pruning two families of pairs

This section is independent of Section 2 and involves no graph: only a finite set with a binary relation. The bound proved here charges the deleted members to two sets of pairs: one determined by the original conflicts, and the other consisting of points left uncovered after deletion.

Lemma 3.1 (Pruning)**.** *Let X be a finite set with a binary relation denoted by $\to$, and let $\mathcal{R}, \mathcal{C} \subseteq X \times X$ be two families, regarded as separate copies even when their coordinates agree. Say that $(p, j) \in \mathcal{R}$ and $(i, s) \in \mathcal{C}$* conflict *if*

$$p \to i \qquad \text{and} \qquad s \to j.$$

Define

$$Z = \{(i, j) : \text{some } (p, j) \in \mathcal{R} \text{ conflicts with some } (i, s) \in \mathcal{C}\},$$
$$H = \{(p, s) : \text{some } (p, j) \in \mathcal{R} \text{ conflicts with some } (i, s) \in \mathcal{C}\}.$$

A point (i, j) is covered *by a member $(p, j) \in \mathcal{R}$ when $p \to i$, and by a member $(i, s) \in \mathcal{C}$ when $s \to j$. These are the only ways a member covers a point.*

There are subfamilies $\mathcal{R}' \subseteq \mathcal{R}$ and $\mathcal{C}' \subseteq \mathcal{C}$ with no conflicts between them such that every initially conflict-free member is retained and

$$|\mathcal{R} \setminus \mathcal{R}'| + |\mathcal{C} \setminus \mathcal{C}'| \leq |H| + |\{z \in Z : z \text{ is covered by no member of } \mathcal{R}' \cup \mathcal{C}'\}|. \tag{13}$$

The two sides are counted separately throughout.

Reading the statement. "Counted separately" means that a pair lying in both $\mathcal{R}$ and $\mathcal{C}$ is two members, one on each side; the left-hand side of (13) counts deletions on each side separately. A member is *initially conflict-free* if it conflicts with no member of the other family. In a conflict the four elements p, i, s, j give the two members $(p, j), (i, s)$ and two further pairs: $(i, j) \in Z$, obtained from (p, j) by advancing the first coordinate along $p \to i$, and $(p, s) \in H$, obtained from (p, j) by moving the second coordinate back along $s \to j$.

Where it is used: once, in Step 8 of the proof of Proposition 4.1.

Remark 3.2 (The lemma is existential, and the last term is needed). Let $X = \{p, s, i_1, i_2, j_1, j_2\}$ with $p \to i_1$, $p \to i_2$, $s \to j_1$, $s \to j_2$, and let $\mathcal{R} = \{(p, j_1), (p, j_2)\}$, $\mathcal{C} = \{(i_1, s), (i_2, s)\}$. Every member of $\mathcal{R}$ conflicts with every member of $\mathcal{C}$, $H = \{(p, s)\}$ and $Z = \{i_1, i_2\} \times \{j_1, j_2\}$. A conflict-free pair must delete one family entirely. Deleting only $\mathcal{C}$ costs 2 and leaves every point of Z covered, so it violates (13) (the right side is 1); deleting everything costs $4 \leq 1 + 4$. So a cheapest pruning need not satisfy the bound, and the bound $|H|$ alone would be false.

We first record the elementary rank fact used in the proof. A real matrix *supported on* a set Θ of positions is one whose entries outside Θ are zero; entries in Θ are allowed to vanish.

Lemma 3.3 (Kernels at maximum support rank)**.** *Let $\mathcal{I}, \mathcal{J}$ be finite sets and $\Theta \subseteq \mathcal{I} \times \mathcal{J}$. Suppose that $W : \mathbb{R}^{\mathcal{J}} \to \mathbb{R}^{\mathcal{I}}$ has maximum rank among all real matrices supported on Θ. Then, for every $\mathbf{x} \in \ker W$, every $\mathbf{y} \in \ker W^{\mathsf{T}}$, and every $(s, j) \in \Theta$, one has $x_j y_s = 0$.*

Where it is used: in Part 8 of the proof of Lemma 3.1, applied to fixed real matrices $W^{p,i}$.

Proof. Suppose $x_j y_s \neq 0$ for some $(s, j) \in \Theta$. Let $W_1 = W + \mathbf{e}_s \mathbf{e}_j^{\mathsf{T}}$, the matrix W with 1 added to its (s, j) entry. It is supported on Θ.

(1) $W_1 \mathbf{x} = W\mathbf{x} + x_j \mathbf{e}_s = x_j \mathbf{e}_s$ and $x_j \neq 0$, so $\mathbf{e}_s \in \operatorname{im} W_1$.

(2) For every $\mathbf{w} \in \mathbb{R}^{\mathcal{J}}$, $W\mathbf{w} = W_1\mathbf{w} - w_j \mathbf{e}_s \in \operatorname{im} W_1$ by (1). So $\operatorname{im} W \subseteq \operatorname{im} W_1$.

(3) $\mathbf{y}^{\mathsf{T}} W = 0$, so $\mathbf{y}$ is orthogonal to $\operatorname{im} W$; but $\mathbf{y}^{\mathsf{T}} \mathbf{e}_s = y_s \neq 0$. So $\mathbf{e}_s \notin \operatorname{im} W$.

By (1)–(3), $\operatorname{im} W_1 \supseteq \operatorname{im} W + \mathbb{R}\mathbf{e}_s \supsetneq \operatorname{im} W$, so $\operatorname{rank} W_1 > \operatorname{rank} W$, contradicting maximality. □

Proof of Lemma 3.1. Plan. Parts 1–2 reduce the lemma to finding a small vertex cover in a bipartite graph Γ. Parts 3–9 show that a maximum matching of Γ has at most $|Z| + |H|$ edges. Part 10 converts the matching bound into the cover (Kőnig's theorem, proved here).

Part 0: trivial case and conventions. If there are no conflicts, retain every member; then $Z = H = \varnothing$ and (13) reads $0 \leq 0$. Assume there is at least one conflict. A 0×0 minor has determinant 1, so the rank arguments below also apply when a matching or an index set is empty.

Part 1: the augmented bipartite graph Γ. Let Z_L and Z_R be two disjoint copies of Z, disjoint from $\mathcal{R}$ and $\mathcal{C}$; write z_L, z_R for the copies of $z \in Z$. Let Γ be the bipartite graph with parts

$$\mathcal{V}_L = \mathcal{R} \sqcup Z_L, \qquad \mathcal{V}_R = \mathcal{C} \sqcup Z_R,$$

and the following edges:

- $\{r, c\}$ for $r \in \mathcal{R}$, $c \in \mathcal{C}$ that conflict;
- $\{r, z_R\}$ for $r \in \mathcal{R}$ and $z \in Z$ covered by r;
- $\{z_L, c\}$ for $c \in \mathcal{C}$ and $z \in Z$ covered by c.

There are no edges between Z_L and Z_R. We call the members of $\mathcal{R} \sqcup \mathcal{C}$ *original* vertices. Figure 1 displays the three edges forced by one conflict.

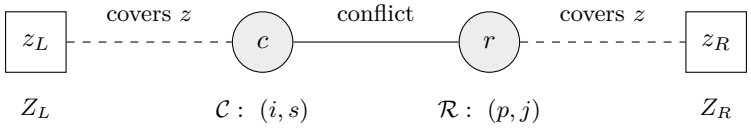


Figure 1: A conflict between $r = (p, j)$ and $c = (i, s)$ forces this path, where $z = (i, j) \in Z$. The copies z_L and z_R are distinct vertices. The vertices alternate between the two sides of the bipartition. Different conflicts can share vertices and edges; the full augmented graph need not be a disjoint union of such paths.

Claim 1: an initially conflict-free member is isolated in Γ. It has no conflict edge by definition. Suppose $(p, j) \in \mathcal{R}$ covers $(i, j) \in Z$, so $p \to i$. Since $(i, j) \in Z$, some $(p', j) \in \mathcal{R}$ conflicts with some $(i, s) \in \mathcal{C}$; in particular $s \to j$. Then $p \to i$ and $s \to j$, so (p, j) conflicts with (i, s). Symmetrically, if $(i, s) \in \mathcal{C}$ covers $(i, j) \in Z$, so $s \to j$, a witness $(p, j) \in \mathcal{R}$, $(i, s') \in \mathcal{C}$ of $(i, j) \in Z$ has $p \to i$, and (p, j) conflicts with (i, s). So a member with a cover edge is not conflict-free.

Part 2: a small vertex cover suffices. *Claim 2: if* K *is a vertex cover of* Γ *with* $|K| \le |Z|+|H|$ *that contains no isolated vertex, then* $\mathcal{R}' = \mathcal{R} \setminus K$ *and* $\mathcal{C}' = \mathcal{C} \setminus K$ *satisfy the lemma.*

No conflict survives, because every conflict edge has an endpoint in K. Conflict-free members are isolated (Claim 1), hence not in K, hence retained. Let q be the number of points of Z covered by no member of $\mathcal{R}' \cup \mathcal{C}'$. Each of the other $|Z| - q$ points z is covered by a surviving member, which is adjacent to z_R (if it is in $\mathcal{R}'$) or to z_L (if it is in $\mathcal{C}'$). That edge must be covered by K, and the surviving member is not in K, so the copy of z is in K. Distinct points have distinct copies, so $|K \cap (Z_L \sqcup Z_R)| \ge |Z| - q$ and

$$|\mathcal{R} \setminus \mathcal{R}'| + |\mathcal{C} \setminus \mathcal{C}'| = |K \cap (\mathcal{R} \sqcup \mathcal{C})| \le |K| - (|Z| - q) \le |H| + q. \tag{14}$$

This is (13).

Part 3: a maximum matching and its three parts. Among all maximum matchings of Γ choose one, $\mathcal{M}$, with the fewest edges joining two original vertices. Split $\mathcal{M} = \mathcal{M}_0 \sqcup \mathcal{M}_\mathcal{R} \sqcup \mathcal{M}_\mathcal{C}$ into its edges between $\mathcal{R}$ and $\mathcal{C}$, between $\mathcal{R}$ and Z_R, and between Z_L and $\mathcal{C}$, and put

$$k = |\mathcal{M}_0|, \qquad \ell = |\mathcal{M}_\mathcal{R}|, \qquad t = |\mathcal{M}_\mathcal{C}|, \qquad |\mathcal{M}| = k + \ell + t.$$

Let $\mathcal{R}_0, \mathcal{R}_1 \subseteq \mathcal{R}$ be the sets of members of $\mathcal{R}$ matched by $\mathcal{M}_0$ and by $\mathcal{M}_\mathcal{R}$, and $\mathcal{C}_0, \mathcal{C}_1 \subseteq \mathcal{C}$ those matched by $\mathcal{M}_0$ and by $\mathcal{M}_\mathcal{C}$. As $\mathcal{M}$ is a matching, $\mathcal{R}_0 \cap \mathcal{R}_1 = \varnothing = \mathcal{C}_0 \cap \mathcal{C}_1$, and $|\mathcal{R}_0| = k$, $|\mathcal{R}_1| = \ell$, $|\mathcal{C}_1| = t$. Let $\Gamma_\mathcal{R}$ be the subgraph of Γ induced on $\mathcal{R} \sqcup Z_R$ and $\Gamma_\mathcal{C}$ the one induced on $Z_L \sqcup \mathcal{C}$.

Claim 3: $\mathcal{M}_\mathcal{R}$ *is a maximum matching of* $\Gamma_\mathcal{R}$*, and* $\mathcal{M}_\mathcal{C}$ *is a maximum matching of* $\Gamma_\mathcal{C}$.

We prove the first assertion; the second is the same with the two sides exchanged. Two facts: (F1) a vertex of Z_R is adjacent only to vertices of $\mathcal{R}$, so it is matched by $\mathcal{M}$ if and only if it is matched by $\mathcal{M}_\mathcal{R}$; (F2) a vertex of $\mathcal{R}$ is adjacent only to vertices of $\mathcal{C} \sqcup Z_R$, so if it is matched by $\mathcal{M}$ but not by $\mathcal{M}_\mathcal{R}$, it is matched by $\mathcal{M}_0$.

Suppose $\Gamma_\mathcal{R}$ has a matching $\mathcal{M}'$ with more than ℓ edges. Consider the graph formed by the edges lying in exactly one of $\mathcal{M}_\mathcal{R}$, $\mathcal{M}'$ (their symmetric difference). Each vertex meets at most one edge of each matching, so it has degree at most 2, and at a vertex of degree 2 the two edges come from different matchings. Hence every component is a path or a cycle whose edges alternate between the two matchings. A cycle, or a path with an even number of edges, has equally many edges of each matching; a path with an odd number of edges has exactly one more edge of the matching that contains its first and last edges. Since $|\mathcal{M}'| > |\mathcal{M}_\mathcal{R}|$, some component is a path π of odd length whose first and last edges are in $\mathcal{M}'$. The two end vertices of π are unmatched by $\mathcal{M}_\mathcal{R}$: if an end vertex x lay on an edge $e' \in \mathcal{M}_\mathcal{R}$, then $e' \notin \mathcal{M}'$ (the only edge of $\mathcal{M}'$ at x is the end edge of π, which is not in $\mathcal{M}_\mathcal{R}$), so e' would be a second edge of the symmetric difference at x. Being of odd length in a bipartite graph, π has one end $r \in \mathcal{R}$ and the other end $z_R \in Z_R$. By (F1), z_R is unmatched in $\mathcal{M}$. Every internal vertex of π is matched in $\mathcal{M}$ by an edge of $\mathcal{M}_\mathcal{R}$ on π, and the other edges of π are not in $\mathcal{M}$ (they are edges of $\Gamma_\mathcal{R}$ outside $\mathcal{M}_\mathcal{R}$).

- If r is unmatched in $\mathcal{M}$, exchanging along π (remove the edges of π in $\mathcal{M}$, add the others) gives a matching of Γ with $|\mathcal{M}| + 1$ edges, contradicting maximality.

- Otherwise, by (F2), r lies on an edge $e \in \mathcal{M}_0$. In $\mathcal{M} \setminus \{e\}$ the vertex r is unmatched, so exchanging along π gives a matching with $(|\mathcal{M}| - 1) + 1 = |\mathcal{M}|$ edges. All edges of π join $\mathcal{R}$ to Z_R, so the exchange adds no edge between two original vertices: the original–original edges of the new matching are $\mathcal{M}_0 \setminus \{e\}$, that is, $k - 1$ of them. This contradicts the choice of $\mathcal{M}$.

Part 4: four matrices with $NL = AB$. For each ordered pair (x, y) with $x \to y$ introduce two variables a_{xy} and b_{xy}, all distinct. For $x \not\to y$ set $a_{xy} = b_{xy} = 0$. For any assignment of real values to the variables, define four matrices

$$\begin{array}{ccc} \mathbb{R}^{\mathcal{R}} & \xrightarrow{L} & \mathbb{R}^{Z} \\ {\scriptstyle B}\downarrow & & \downarrow{\scriptstyle N} \\ \mathbb{R}^{H} & \xrightarrow{A} & \mathbb{R}^{\mathcal{C}} \end{array}$$

with entries

$$\begin{aligned} L_{(i,j),(p,j)} &= a_{pi}, \quad N_{(i,s),(i,j)} = b_{sj}, \\ B_{(p,s),(p,j)} &= b_{sj}, \quad A_{(i,s),(p,s)} = a_{pi}. \end{aligned} \tag{15}$$

In each formula the row index is written first and ranges over the target's index set, the column index over the source's. Every entry not of a listed form is zero: an entry of L or A vanishes unless row and column have the *same second coordinate*, and an entry of N or B vanishes unless they have the *same first coordinate*. So L, A advance the first coordinate along $p \to i$, and N, B move the second coordinate back from j to s along $s \to j$. Two consequences of the convention $a_{xy} = b_{xy} = 0$ for $x \not\to y$:

(O1) $L_{z,r} \neq 0$ only if $r \in \mathcal{R}$ covers z, that is, only if $\{r, z_R\}$ is an edge of $\Gamma_{\mathcal{R}}$;

(O2) $N_{c,z} \neq 0$ only if $c \in \mathcal{C}$ covers z, that is, only if $\{z_L, c\}$ is an edge of $\Gamma_{\mathcal{C}}$.

Claim 4: for every assignment,

$$NL = AB. \tag{16}$$

Fix a column $(p, j) \in \mathcal{R}$ and a row $(i, s) \in \mathcal{C}$. The product NL passes through $\mathbb{R}^Z$, so $(NL)_{(i,s),(p,j)} = \sum_{z \in Z} N_{(i,s),z} L_{z,(p,j)}$. A term can be nonzero only if z has the first coordinate i of the row (form of N) and the second coordinate j of the column (form of L), that is, $z = (i, j)$. So the entry is $b_{sj} a_{pi}$ if $(i, j) \in Z$, and 0 if $(i, j) \notin Z$ (no term is left). Likewise AB passes through $\mathbb{R}^H$, only the index (p, s) can contribute, and $(AB)_{(i,s),(p,j)}$ is $a_{pi} b_{sj}$ if $(p, s) \in H$ and 0 otherwise. If (p, j) and (i, s) conflict, then $(i, j) \in Z$ and $(p, s) \in H$ by definition, and both entries equal $a_{pi} b_{sj}$. If they do not conflict, then $p \not\to i$ or $s \not\to j$, so $a_{pi} = 0$ or $b_{sj} = 0$ by convention; each of the two entries is then either 0 for lack of a term or equal to $a_{pi} b_{sj} = 0$, whether or not $(i, j) \in Z$ and $(p, s) \in H$.

Part 5: three rank conditions imply the matching bound. *Claim 5: if for some real assignment*

$$\operatorname{rank} L = \ell, \qquad \operatorname{rank} N = t, \qquad \dim \ker A \geq k, \tag{17}$$

then $|\mathcal{M}| = k + \ell + t \leq |Z| + |H|$. Indeed, by rank–nullity for $A : \mathbb{R}^H \to \mathbb{R}^{\mathcal{C}}$, then $\operatorname{rank} A \geq \operatorname{rank}(AB)$, then (16),

$$|H| = \dim \ker A + \operatorname{rank} A \geq k + \operatorname{rank}(AB) = k + \operatorname{rank}(NL) \geq k + \ell + t - |Z|. \tag{18}$$

The last inequality is rank–nullity for N restricted to $\operatorname{im} L$:

$$\operatorname{rank}(NL) = \dim \operatorname{im} L - \dim(\operatorname{im} L \cap \ker N) \geq \ell - \dim \ker N = \ell - (|Z| - t).$$

Parts 6–9 produce an assignment satisfying (17).

Part 6: one assignment satisfying finitely many genericity conditions. The correspondence between generic matrix rank and maximum matching size is due to Edmonds [9, Section 5, Theorem 1]. The citation is for attribution only; the two facts used are proved below: for every real assignment the rank is at most the maximum matching size (6a), and the minor on the endpoints of the chosen matching is a nonzero polynomial (6b), (6c). Let $\mathbb{R}[a, b]$ be the polynomial ring in all the variables a_{xy}, b_{xy} $(x \to y)$. It is an integral domain, so a product of nonzero polynomials is nonzero. The entries of L and N, and of the matrices $W^{p,i}$ defined below, are variables or 0, so each minor of these matrices is an element of $\mathbb{R}[a, b]$, and its value at a real assignment is the corresponding minor of the real matrix.

(6a) $\operatorname{rank} L \leq \ell$ *and* $\operatorname{rank} N \leq t$ *for every real assignment.* Suppose L has a nonzero $h \times h$ minor, with column set $\mathcal{R}^\circ \subseteq \mathcal{R}$ and row set $Z^\circ \subseteq Z$. Some term of the Leibniz expansion is nonzero: there is a bijection $\varphi : \mathcal{R}^\circ \to Z^\circ$ with $L_{\varphi(r),r} \neq 0$ for all $r \in \mathcal{R}^\circ$. By (O1) the edges $\{r, \varphi(r)_R\}$ form a matching of $\Gamma_{\mathcal{R}}$ with h edges, so $h \leq \ell$ by Claim 3. The same argument with (O2) and $\Gamma_{\mathcal{C}}$ gives $\operatorname{rank} N \leq t$.

(6b) The matching minor π_L *of* L *is a nonzero polynomial.* Let $Z' \subseteq Z$ be the set of points z whose right copy z_R is matched by $\mathcal{M}_{\mathcal{R}}$, and let $\varphi_0 : \mathcal{R}_1 \to Z'$ be the bijection given by $\mathcal{M}_{\mathcal{R}}$. Let $\pi_L \in \mathbb{R}[a, b]$ be the determinant of the $\ell \times \ell$ submatrix of L with columns $\mathcal{R}_1$ and rows Z'. An edge $\{r, z_R\}$ means that $r = (p, j)$ covers $z = (i, j)$: the second coordinates agree and $p \to i$. So φ_0 preserves second coordinates. Group the columns in $\mathcal{R}_1$ and the rows in Z' by their second coordinate j; for each j the two groups have the same size (via φ_0), and entries of L between different groups vanish. So the submatrix is block diagonal up to reordering, and $\pi_L = \pm \prod_j \pi_{L,j}$, where $\pi_{L,j}$ is the determinant of the block for j.

Fix j. The block has columns (p, j), $p \in P^j$, rows (i, j), $i \in I^j$, and entry a_{pi} in position $((i, j), (p, j))$. Its determinant is

$$\pi_{L,j} = \sum_{\sigma} \pm \prod_{p \in P^j} a_{p\,\sigma(p)},$$

summed over bijections $\sigma : P^j \to I^j$; the term of σ is nonzero exactly when $p \to \sigma(p)$ for all p, and then it is a product of $|P^j|$ *distinct* variables. This monomial determines σ: the variable a_{pi} occurs in it if and only if $\sigma(p) = i$. Hence different bijections give different monomials and nothing cancels. The bijection induced by φ_0 satisfies $p \to \sigma(p)$ for all p, so its monomial occurs with coefficient ± 1. Thus $\pi_{L,j} \neq 0$ in $\mathbb{R}[a, b]$. The same variable a_{pi} may occur in blocks for different j; this is why we argue block by block and then use that $\mathbb{R}[a, b]$ is an integral domain to conclude $\pi_L \neq 0$.

(6c) The matching minor π_N *of* N *is a nonzero polynomial.* Let $Z'' \subseteq Z$ be the set of points z whose left copy z_L is matched by $\mathcal{M}_{\mathcal{C}}$, and π_N the determinant of the $t \times t$ submatrix of N with rows $\mathcal{C}_1$ and columns Z''. An edge $\{z_L, c\}$ means that $c = (i, s)$ covers $z = (i, j)$: the first coordinates agree and $s \to j$. Repeating (6b) with blocks of constant *first* coordinate i, in which the entries are distinct variables b_{sj}, gives $\pi_N \neq 0$.

(6d) The auxiliary matrices $W^{p,i}$. For every pair with $p \to i$ define

$$J_p = \{j : (p, j) \in \mathcal{R}\}, \qquad \Sigma_i = \{s : (i, s) \in \mathcal{C}\}, \qquad W^{p,i} = (b_{sj})_{s \in \Sigma_i,\ j \in J_p},$$

a matrix $\mathbb{R}^{J_p} \to \mathbb{R}^{\Sigma_i}$, and

$$\Theta^{p,i} = \{(s,j) \in \Sigma_i \times J_p : s \to j\}.$$

The entries of $W^{p,i}$ in positions of $\Theta^{p,i}$ are pairwise distinct variables, and the other entries are 0. Hence at any real assignment $W^{p,i}$ is a real matrix supported on $\Theta^{p,i}$, and every real matrix supported on $\Theta^{p,i}$ arises from some assignment. Let $\kappa_{p,i}$ be the maximum rank of a real matrix supported on $\Theta^{p,i}$, attained by W_0 say. Some $\kappa_{p,i} \times \kappa_{p,i}$ minor of W_0 is nonzero; let $\pi_{p,i} \in \mathbb{R}[a,b]$ be the minor of $W^{p,i}$ on the same rows and columns (the empty minor 1 if $\kappa_{p,i} = 0$). The assignment $b_{sj} := (W_0)_{sj}$ for $(s,j) \in \Theta^{p,i}$, with arbitrary values for the other variables, turns $W^{p,i}$ into W_0 and $\pi_{p,i}$ into a nonzero number. So $\pi_{p,i} \neq 0$ in $\mathbb{R}[a,b]$.

(6e) A common non-root. The product

$$\pi = \pi_L \cdot \pi_N \cdot \prod_{p \to i} \pi_{p,i}$$

has finitely many factors, each a nonzero element of $\mathbb{R}[a,b]$, so $\pi \neq 0$. It does not matter that the same variables b_{sj} occur in N, in B and in several matrices $W^{p,i}$: we need only one point where the single polynomial π does not vanish. A nonzero real polynomial is nonzero at some real point (induction on the number of variables: write it as a polynomial in the last variable; some coefficient is a nonzero polynomial in the remaining variables and is nonzero at some point by induction; the resulting nonzero one-variable polynomial has finitely many roots). *Fix, once and for all, a real assignment with $\pi \neq 0$.* All matrices below are the real matrices at this assignment.

(6f) What the assignment gives.

(G1) $\pi_L \neq 0$: the ℓ columns of L indexed by $\mathcal{R}_1$ are linearly independent. With (6a), $\operatorname{rank} L = \ell$ and these columns are a basis of $\operatorname{im} L$.

(G2) $\pi_N \neq 0$: the t rows of N indexed by $\mathcal{C}_1$, that is, the columns of N^{T} indexed by $\mathcal{C}_1$, are linearly independent. With (6a), $\operatorname{rank} N = \operatorname{rank} N^{\mathsf{T}} = t$ and these columns are a basis of $\operatorname{im} N^{\mathsf{T}}$.

(G3) $\pi_{p,i} \neq 0$: the real matrix $W^{p,i}$ has a nonzero $\kappa_{p,i} \times \kappa_{p,i}$ minor, so $\operatorname{rank} W^{p,i} \geq \kappa_{p,i}$. By (6d) it is itself a real matrix supported on $\Theta^{p,i}$, so $\operatorname{rank} W^{p,i} \leq \kappa_{p,i}$ by the definition of $\kappa_{p,i}$. Hence, for every $p \to i$, $W^{p,i}$ has maximum rank among all real matrices supported on $\Theta^{p,i}$.

In particular the first two conditions of (17) hold.

Part 7: lifting the original–original edges into kernels. Let $\{r,c\} \in \mathcal{M}_0$, so $r \in \mathcal{R}_0$, $c \in \mathcal{C}_0$. By (G1), $L\mathbf{e}_r$ is a combination of the columns $L\mathbf{e}_{r'}$, $r' \in \mathcal{R}_1$; by (G2), $N^{\mathsf{T}}\mathbf{e}_c$ is a combination of the columns $N^{\mathsf{T}}\mathbf{e}_{c'}$, $c' \in \mathcal{C}_1$. So there are real numbers $\lambda^r_{r'}$, $\mu^c_{c'}$ with

$$\begin{aligned} \mathbf{u}^r &= \mathbf{e}_r + \sum_{r' \in \mathcal{R}_1} \lambda^r_{r'} \mathbf{e}_{r'} \in \ker L, \\ \mathbf{v}^c &= \mathbf{e}_c + \sum_{c' \in \mathcal{C}_1} \mu^c_{c'} \mathbf{e}_{c'} \in \ker N^{\mathsf{T}}. \end{aligned} \tag{19}$$

Since $\mathcal{R}_0 \cap \mathcal{R}_1 = \varnothing$ and $\mathcal{C}_0 \cap \mathcal{C}_1 = \varnothing$,

$$u^r_r = 1, \qquad u^r_{r''} = 0 \ \text{ for } r'' \in \mathcal{R}_0 \setminus \{r\}, \qquad v^c_c = 1. \tag{20}$$

Part 8: the kernel restriction. *Claim 8: at the fixed assignment,*

$$u_{(p,j)}\, v_{(i,s)} = 0 \quad \text{for every conflict } ((p,j),(i,s)), \text{ every } \mathbf{u} \in \ker B, \text{ and every } \mathbf{v} \in \ker N^{\mathsf{T}}. \tag{21}$$

Fix a conflict, so $p \to i$, $j \in J_p$, $s \in \Sigma_i$ and $(s,j) \in \Theta^{p,i}$; fix $\mathbf{u} \in \ker B$ and $\mathbf{v} \in \ker N^{\mathsf{T}}$. Put $\mathbf{x} = (u_{(p,j')})_{j' \in J_p}$ and $\mathbf{y} = (v_{(i,s')})_{s' \in \Sigma_i}$.

(8a) $\mathbf{x} \in \ker W^{p,i}$. We must show, for every row $s' \in \Sigma_i$,

$$\sum_{j' \in J_p} b_{s'j'}\, u_{(p,j')} = 0. \tag{22}$$

If $s' \to j'$ for no $j' \in J_p$, all coefficients are 0 and the equation is trivially true. Otherwise pick $j' \in J_p$ with $s' \to j'$; then $(p,j') \in \mathcal{R}$ and $(i,s') \in \mathcal{C}$ conflict (as $p \to i$), so $(p,s') \in H$. By (15), the row of B indexed by (p,s') has entry $b_{s'j'}$ in each column (p,j'), $j' \in J_p$, and 0 in all other columns; so (22) is precisely the (p,s') coordinate of $B\mathbf{u} = 0$. (The matrix B has a row (p,s') only when $(p,s') \in H$. By the previous sentences, $(p,s') \notin H$ can happen only in the trivial case, so the two cases together cover every row $s' \in \Sigma_i$.)

(8b) $\mathbf{y} \in \ker(W^{p,i})^{\mathsf{T}}$. We must show, for every column $j' \in J_p$,

$$\sum_{s' \in \Sigma_i} b_{s'j'}\, v_{(i,s')} = 0. \tag{23}$$

If $s' \to j'$ for no $s' \in \Sigma_i$, the equation is trivial. Otherwise a conflict between (p,j') and some (i,s') gives $(i,j') \in Z$. The column of N indexed by (i,j') has entry $b_{s'j'}$ in each row (i,s'), $s' \in \Sigma_i$, and 0 elsewhere; so (23) is the (i,j') coordinate of $N^{\mathsf{T}}\mathbf{v} = 0$.

(8c) Conclusion. By (G3), the real matrix $W^{p,i}$ has maximum rank among real matrices supported on $\Theta^{p,i}$. Lemma 3.3, applied to this one fixed real matrix with $\mathbf{x}, \mathbf{y}$ and $(s,j) \in \Theta^{p,i}$, gives $x_j y_s = u_{(p,j)} v_{(i,s)} = 0$. The perturbed matrix W_1 inside the proof of Lemma 3.3 is only a device in that proof: the assignment fixed in (6e) is not changed, and no other rank condition is affected.

Part 9: $\dim\ker A \geq k$. *(9a) Every* $\mathbf{u} \in \ker B$ *vanishes on* $\mathcal{R}_0$. Let $r \in \mathcal{R}_0$ with mate c, so $\{r,c\} \in \mathcal{M}_0$. An edge of Γ between original vertices is a conflict, so (21) with $\mathbf{v} = \mathbf{v}^c$ and $v^c_c = 1$ gives $u_r = 0$.

(9b) The k *vectors* $B\mathbf{u}^r$, $r \in \mathcal{R}_0$, *are linearly independent.* Suppose $\sum_{r \in \mathcal{R}_0} \theta_r B\mathbf{u}^r = 0$. Then $\mathbf{u} = \sum_r \theta_r \mathbf{u}^r \in \ker B$. By (20) its coordinate at each $r \in \mathcal{R}_0$ is θ_r, and by (9a) it is 0.

(9c) They lie in $\ker A$. By (16) and (19), $A(B\mathbf{u}^r) = N(L\mathbf{u}^r) = 0$.

Hence $\dim\ker A \geq k$. Together with (G1), (G2) all of (17) holds, and Claim 5 gives

$$|\mathcal{M}| = k + \ell + t \leq |Z| + |H|. \tag{24}$$

Part 10: from the matching bound to a vertex cover. For completeness, we give the matching-to-cover argument in Kőnig's theorem [17]. Starting at all vertices of $\mathcal{V}_L$ unmatched by $\mathcal{M}$, follow edges outside $\mathcal{M}$ from left to right and edges in $\mathcal{M}$ from right to left. Let $\mathcal{V}_L^\star, \mathcal{V}_R^\star$ be the sets of reached vertices on the two sides (the starting vertices are reached).

(i) No unmatched vertex of $\mathcal{V}_R$ is reached: an alternating path to one would be an augmenting path, contradicting maximality of $\mathcal{M}$.

(ii) The two endpoints of an edge of $\mathcal{M}$ are both reached or both unreached: a reached right endpoint leads to its mate, and a matched left vertex is not a starting vertex, so it can only be reached through its mate.

(iii) $K = (\mathcal{V}_L \setminus \mathcal{V}_L^\star) \cup \mathcal{V}_R^\star$ is a vertex cover. Otherwise some edge joins a reached left vertex to an unreached right vertex. It is not outside $\mathcal{M}$ (it would be traversed), and not in $\mathcal{M}$ by (ii).

(iv) $|K| = |\mathcal{M}|$ and K contains no isolated vertex. Unmatched left vertices are reached and unmatched right vertices are unreached by (i), so K consists of matched vertices. By (ii), K contains exactly one endpoint of each edge of $\mathcal{M}$: the right one if both are reached, the left one otherwise. Isolated vertices are unmatched.

By (iv) and (24), $|K| \le |Z| + |H|$, and Claim 2 applied to K proves the lemma. □

4 The extremal contradiction

We use Lemma 3.1 to rule out the graph supplied by Proposition 2.1.

Proposition 4.1. *There is no nonempty finite oriented graph G in which every vertex has an in-neighbor and whose one-step image operator*

$$F(U) = \{y \in V(G) : x \to y \text{ for some } x \in U\}$$

satisfies

$$|U| + |F^2(U)| < 2|F(U)| \qquad (\varnothing \neq U \subsetneq V(G)), \tag{25}$$

where $F^2 = F \circ F$.

Plan of the proof. Assume G exists. We choose an extremal compatible pair (P, Q) of families of ordered pairs (Steps 1–2), use (25) on columns and rows to build larger but possibly conflicting families $\mathcal{R}, \mathcal{C}$ (Steps 3–4), check the three facts needed to apply the pruning lemma profitably (Steps 5–7), and compare objective values (Step 8).

Proof. Suppose that such a graph exists, and put $X = V(G)$; throughout, $\to$ is the arc relation of G and $F = F_G$.

Step 0: coordinate images and compatibility. For a set $\mathcal{S} \subseteq X \times X$ of ordered pairs define

$$F_1\mathcal{S} = \{(i, j) : (p, j) \in \mathcal{S} \text{ and } p \to i \text{ for some } p \in X\},$$
$$F_2\mathcal{S} = \{(i, j) : (i, s) \in \mathcal{S} \text{ and } s \to j \text{ for some } s \in X\}.$$

Write $F_1^2 = F_1 \circ F_1$, $F_2^2 = F_2 \circ F_2$, and $\Delta = \{(v, v) : v \in X\}$. Call $P, Q \subseteq X \times X$ *compatible* if $F_1P \cap F_2Q = \varnothing$. In the language of Lemma 3.1, with P as the left family and Q as the right family: F_1P is exactly the set of points covered by some member of P, F_2Q is the set of points covered by some member of Q, and a point (i, j) lies in $F_1P \cap F_2Q$ if and only if some $(p, j) \in P$ and some $(i, s) \in Q$ satisfy $p \to i$ and $s \to j$. Hence

$$P, Q \text{ compatible} \iff \text{no member of } P \text{ conflicts with a member of } Q. \tag{26}$$

For a compatible pair put

$$d(P, Q) = |X|^2 - |F_1P| - |F_2Q|, \qquad \Phi(P, Q) = |P| + |Q| + d(P, Q).$$

Since the two images are disjoint, $d(P, Q)$ is the number of points of $X \times X$ outside $F_1P \cup F_2Q$, the *uncovered* points.

Step 1: an extremal pair. The pair (Δ, Δ) is compatible: $(i, j) \in F_1\Delta$ means $j \to i$, and $(i, j) \in F_2\Delta$ means $i \to j$; both together are impossible in an oriented graph (opposite arcs if $i \neq j$,

a loop if $i = j$). So the set of compatible pairs (P, Q) with $\Delta \subseteq P \cap Q$ is nonempty and finite. Choose one maximizing Φ, and write $d = d(P, Q)$.

Step 2: columns of P and rows of Q are nonempty proper subsets. For $j \in X$ let $P_j = \{p : (p, j) \in P\}$; it contains j. Suppose $P_j = X$. Positive indegrees give $F(X) = X$, so $X \times \{j\} \subseteq F_1 P$. Choose an in-neighbour x of j. Then $(x, x) \in \Delta$ and $x \to j$ give $(x, j) \in F_2\Delta \subseteq F_2 Q$, contradicting compatibility. For $i \in X$ let $Q_i = \{s : (i, s) \in Q\}$; it contains i. If $Q_i = X$, then $\{i\} \times X \subseteq F_2 Q$; choosing $x \to i$ gives $(i, x) \in F_1\Delta \subseteq F_1 P$, again a contradiction.

Step 3: summing the strict inequality over fibres. The column of $F_1 P$ over j is $F(P_j) \times \{j\}$ and that of $F_1^2 P$ is $F^2(P_j) \times \{j\}$. By Step 2, (25) applies to each P_j; summing over the $|X| \geq 1$ values of j gives the first inequality below. The second is the same with the rows Q_i, since the row of $F_2 Q$ at i is $\{i\} \times F(Q_i)$:

$$|P| + |F_1^2 P| < 2|F_1 P|, \qquad |Q| + |F_2^2 Q| < 2|F_2 Q|. \tag{27}$$

Step 4: the complement families and their size. Define

$$\mathcal{R} = (X \times X) \setminus F_2^2 Q, \qquad \mathcal{C} = (X \times X) \setminus F_1^2 P.$$

(The left family comes from Q and the right family from P.) By (27),

$$\begin{aligned} |\mathcal{R}| + |\mathcal{C}| &= 2|X|^2 - |F_2^2 Q| - |F_1^2 P| \\ &> 2|X|^2 + |P| + |Q| - 2(|F_1 P| + |F_2 Q|) = |P| + |Q| + 2d. \end{aligned} \tag{28}$$

These families need not be compatible.

Step 5: $\Delta \subseteq \mathcal{R} \cap \mathcal{C}$. Let $v \in X$. If $(v, v) \in F_2^2 Q$, there is $s \to v$ with $(v, s) \in F_2 Q$. But $(s, s) \in P$ and $s \to v$ give $(v, s) \in F_1 P$, contradicting compatibility of (P, Q). If $(v, v) \in F_1^2 P$, there is $p \to v$ with $(p, v) \in F_1 P$. But $(p, p) \in Q$ and $p \to v$ give $(p, v) \in F_2 Q$, the same contradiction.

Step 6: diagonal members are conflict-free in $(\mathcal{R}, \mathcal{C})$. Suppose the left member $(v, v) \in \mathcal{R}$ conflicts with a right member $(i, s) \in \mathcal{C}$: $v \to i$ and $s \to v$. From $(s, s) \in P$ and $s \to v$ we get $(v, s) \in F_1 P$, and then $v \to i$ gives $(i, s) \in F_1^2 P$, contradicting $(i, s) \in \mathcal{C}$. Suppose a left member $(p, j) \in \mathcal{R}$ conflicts with the right member $(v, v) \in \mathcal{C}$: $p \to v$ and $v \to j$. From $(p, p) \in Q$ and $p \to v$ we get $(p, v) \in F_2 Q$, and then $v \to j$ gives $(p, j) \in F_2^2 Q$, contradicting $(p, j) \in \mathcal{R}$.

Step 7: $|H| \leq d$. Let $H, Z \subseteq X \times X$ be the sets associated with $\mathcal{R}, \mathcal{C}$ in Lemma 3.1: each conflict

$$(p, j) \in \mathcal{R}, \quad (i, s) \in \mathcal{C}, \quad p \to i, \quad s \to j$$

contributes (p, s) to H and (i, j) to Z. We claim that

$$H \subseteq (X \times X) \setminus (F_1 P \cup F_2 Q), \qquad \text{and hence} \qquad |H| \leq d. \tag{29}$$

Fix such a conflict. If $(p, s) \in F_1 P$, then $p \to i$ gives $(i, s) \in F_1^2 P$, contrary to $(i, s) \in \mathcal{C}$. If $(p, s) \in F_2 Q$, then $s \to j$ gives $(p, j) \in F_2^2 Q$, contrary to $(p, j) \in \mathcal{R}$. So (p, s) is one of the d uncovered points of (P, Q).

Step 8: pruning and the contradiction. Apply Lemma 3.1 to X, the relation $\to$, and $\mathcal{R}, \mathcal{C}$. We obtain $P' \subseteq \mathcal{R}$ and $Q' \subseteq \mathcal{C}$ without conflicts between them, retaining all conflict-free members, with

$$|\mathcal{R} \setminus P'| + |\mathcal{C} \setminus Q'| \leq |H| + q, \tag{30}$$

where q is the number of points of Z covered by no member of $P' \cup Q'$.

- $\Delta \subseteq P' \cap Q'$, by Steps 5 and 6 and the retention clause.
- (P', Q') is compatible, by (26). So it is one of the pairs considered in Step 1, and $\Phi(P', Q') \leq \Phi(P, Q)$.
- Put $d' = d(P', Q')$. By Step 0, a point is covered by a member of the left family P' exactly when it lies in $F_1 P'$, and by a member of the right family Q' exactly when it lies in $F_2 Q'$. So the q uncovered points of Z are among the d' points of $X \times X$ outside $F_1 P' \cup F_2 Q'$: $q \leq d'$.

By (30), (29) and $q \leq d'$, at most $d + d'$ members were deleted. Using (28),

$$\begin{aligned}\Phi(P', Q') = |P'| + |Q'| + d' &\geq |\mathcal{R}| + |\mathcal{C}| - (d + d') + d' \\ &> |P| + |Q| + 2d - (d + d') + d' = |P| + |Q| + d = \Phi(P, Q).\end{aligned}$$

This contradicts the maximality of $\Phi(P, Q)$ and proves the proposition. □

Proof of Theorem 1.1. If the theorem were false, Proposition 2.1 would produce a nonempty finite oriented graph with positive indegrees satisfying (25) for every nonempty proper vertex set ((3) and (25) are the same inequality). Proposition 4.1 rules out such a graph. □

5 Weighted consequences

The following corollary applies Theorem 1.1 through Seacrest's established equivalences of the corresponding assertions over all nonempty finite oriented graphs [19]. The equivalences themselves are not new here. The two vertex-weighted conclusions have different quantifier orders: in (i), for every weighting there is a suitable vertex; in (ii), there is one weighting for which every vertex is suitable.

Corollary 5.1 (Weighted second-neighborhood forms)**.** *Let D be a nonempty finite oriented graph, and write* $\mathrm{Arc}(D)$ *for its arc set. For a vertex weighting $\eta : V(D) \to [0, \infty)$, write $\eta(S) = \sum_{x \in S} \eta(x)$. Then the following hold.*

(i) *For every vertex weighting $\eta : V(D) \to [0, \infty)$, there is a vertex $v \in V(D)$ such that*

$$\eta(N_1^+(v)) \leq \eta(N_2^+(v)).$$

(ii) *There is a vertex weighting $\eta : V(D) \to [0, \infty)$ with $\sum_{x \in V(D)} \eta(x) = 1$ such that*

$$\eta(N_1^+(v)) \leq \eta(N_2^+(v)) \qquad \text{for every } v \in V(D).$$

The weighting is nonnegative; some of its entries may be zero.

(iii) *For every arc weighting $w : \mathrm{Arc}(D) \to [0, \infty)$, extend the notation by setting $w(vs) = 0$ when $vs \notin \mathrm{Arc}(D)$. For $v, s \in V(D)$, put*

$$\beta_v^w(s) = \max\Big(\{0\} \cup \{w(us) - w(vs) : u \in V(D),\ v \to u \to s\}\Big).$$

Then there is a vertex $v \in V(D)$ such that

$$\sum_{s \in V(D)} \beta_v^w(s) \geq \sum_{u \in N_1^+(v)} w(vu).$$

Conventions in (iii). The condition $v \to u \to s$ refers to arcs of D and does not depend on w: an arc of weight zero is still an arc. The number $w(vs)$ enters only as a value, equal to 0 when vs is not an arc. If there is no path $v \to u \to s$, then $\beta_v^w(s) = 0$.

Proof. (iii). As in the source, this part is obtained by citation and is not reproved. The cited result is [19, Proposition 2]: Seacrest's arc-weighted Conjecture 4, which asserts the conclusion of (iii) for all nonempty finite oriented graphs and all nonnegative arc weightings, is equivalent to the unweighted conjecture. Theorem 1.1 proves the unweighted conjecture, so (iii) follows. A reader who prefers not to rely on the reference can use Remark 5.2, an optional direct argument for the one implication used; it was added in this rewrite and plays no other role.

(i) *from* (iii). Given η, set $w(us) = \eta(s)$ on every arc us. We claim that for all v, s,

$$\beta_v^w(s) = \begin{cases} \eta(s), & s \in N_2^+(v), \\ 0, & s \notin N_2^+(v). \end{cases}$$

If there is no path $v \to u \to s$ (this includes $s = v$), then $\beta_v^w(s) = 0$ and $s \notin N_2^+(v)$. Otherwise every such u has $w(us) = \eta(s)$. If $v \to s$, then $w(vs) = \eta(s)$, all differences are 0, and $s \notin N_2^+(v)$. If $v \not\to s$, then $s \in N_2^+(v)$, $w(vs) = 0$ and the maximum is $\eta(s)$. Thus the left side of (iii) is $\eta(N_2^+(v))$ and the right side is $\sum_{u \in N_1^+(v)} \eta(u) = \eta(N_1^+(v))$. So (i) follows from (iii). (The source adds that Seacrest's paper notes this specialization immediately after his Proposition 2; the computation just given is complete without that note.)

(ii) *from* (i). Index rows and columns by $V(D)$ and let Π_D be the matrix with entry 1 at (v, s) when $s \in N_2^+(v)$, entry -1 when $s \in N_1^+(v)$, and zero otherwise. For $\eta \geq 0$ the v-th coordinate of $\Pi_D \eta$ is $\eta(N_2^+(v)) - \eta(N_1^+(v))$. So (ii) asks for $\eta \geq 0$, $\eta \neq 0$, with $\Pi_D \eta \geq 0$, normalized to total weight one.

Step 1: reversal transposes. Let $\overleftarrow{D}$ be D with all arcs reversed; it is an oriented graph. Then s is an out-neighbour of v in $\overleftarrow{D}$ iff $v \in N_1^+(s)$ in D, and there is a path from v to s of length two in $\overleftarrow{D}$ iff there is one from s to v in D. Hence s is an exact second out-neighbour of v in $\overleftarrow{D}$ iff $v \in N_2^+(s)$ in D, and $\Pi_{\overleftarrow{D}} = \Pi_D^{\mathsf{T}}$.

Step 2: an alternative. The source attributes the following to [19, Theorem 1], which applies Farkas' lemma: for a real square matrix Π, at least one of

(A1) there is $\eta \geq 0$, $\eta \neq 0$, with $\Pi\eta \geq 0$;

(A2) there is $\xi \geq 0$ with $\Pi^{\mathsf{T}}\xi < 0$ in every coordinate

holds. For the reader's convenience, here is the derivation from Farkas' lemma in the form: *for a real matrix M_0 and vector $\mathbf{c}$, either $M_0\mathbf{x} = \mathbf{c}$ has a solution $\mathbf{x} \geq 0$, or there is $\mathbf{y}_0$ with $M_0^{\mathsf{T}}\mathbf{y}_0 \geq 0$ and $\mathbf{c}^{\mathsf{T}}\mathbf{y}_0 < 0$.* Take, in block form (with $\mathbf{1}$ the all-ones vector and Id the identity matrix),

$$M_0 = \begin{pmatrix} \Pi & -\mathrm{Id} \\ \mathbf{1}^{\mathsf{T}} & 0 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} \eta \\ \sigma \end{pmatrix}, \qquad \mathbf{c} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then $M_0\mathbf{x} = \mathbf{c}$, $\mathbf{x} \geq 0$ says $\eta \geq 0$, $\sigma = \Pi\eta \geq 0$ and $\mathbf{1}^{\mathsf{T}}\eta = 1$; this is (A1) after scaling η to total weight one. If there is no solution, Farkas' lemma gives $\mathbf{y}_0 = (\mathbf{y}, \tau)$, with $\mathbf{y}$ a vector and τ a number, such that

$$M_0^{\mathsf{T}}\mathbf{y}_0 = \begin{pmatrix} \Pi^{\mathsf{T}}\mathbf{y} + \tau\mathbf{1} \\ -\mathbf{y} \end{pmatrix} \geq 0 \qquad \text{and} \qquad \mathbf{c}^{\mathsf{T}}\mathbf{y}_0 = \tau < 0.$$

Put $\xi = -\mathbf{y}$. The second block gives $\xi \geq 0$, and the first gives $\Pi^{\mathsf{T}}\xi = -\Pi^{\mathsf{T}}\mathbf{y} \leq \tau\mathbf{1} < 0$ in every coordinate, which is (A2).

Step 3: conclusion. Apply Step 2 to $\Pi = \Pi_D$. If (A2) held, then by Step 1 $\Pi_{\overleftarrow{D}}\,\xi < 0$ in every coordinate, that is, $\xi(N_1^+(v)) > \xi(N_2^+(v))$ for every vertex v of $\overleftarrow{D}$. This contradicts (i) applied to $\overleftarrow{D}$ and the weighting ξ. So (A1) holds; dividing η by its total weight proves (ii). □

Remark 5.2 (Optional direct check of (iii); added in this rewrite, not in the source). Only the implication "Theorem 1.1 ⇒ (iii)" is used, and it can be checked without the reference.

Integer weights. Let w be integer-valued and $w_{\max} = \max(1, \max w)$. Let D' have vertex set $V(D) \times \{1, \dots, w_{\max}\}$ and an arc $(u, b) \to (s, c)$ exactly when $us \in \mathrm{Arc}(D)$ and $c \leq w(us)$. A loop or a pair of opposite arcs in D' would project to one in D, so D' is a nonempty oriented graph. Fix a vertex (v, a) of D'. The set $N_1^+((v,a)) = \{(u,c) : vu \in \mathrm{Arc}(D),\ c \leq w(vu)\}$ has $\sum_{u \in N_1^+(v)} w(vu)$ elements. Let $(s,c) \in N_2^+((v,a))$. There is a path $(v,a) \to (u,b) \to (s,c)$, so $v \to u \to s$ in D and $c \leq w(us)$. Also $(s,c) \notin N_1^+((v,a))$, which means $vs \notin \mathrm{Arc}(D)$ or $c > w(vs)$; in the first case $w(vs) = 0 < c$ by the convention of (iii), so $c > w(vs)$ in both cases. Hence, for fixed s, every such c is an integer with $w(vs) < c \leq \max_u w(us)$, the maximum being over the u with $v \to u \to s$. There are $\max(0, \max_u w(us) - w(vs)) = \beta_v^w(s)$ such integers (and no such c if there is no such u). So $|N_2^+((v,a))| \leq \sum_s \beta_v^w(s)$, and the vertex (v,a) given by Theorem 1.1 for D' yields (iii) for v.

Rational weights. Choose a positive integer γ with γw integer-valued. Replacing w by γw multiplies both sides of (iii) by γ, so the integer case gives the rational case.

Real weights. Choose nonnegative rational weightings $w_1, w_2, \dots$ converging to w on every arc, and for each k a vertex v_k satisfying (iii) for w_k. As $V(D)$ is finite, some vertex v equals v_k for infinitely many k. For this fixed v, the right side of (iii) is linear in w, and each $\beta_v^w(s)$ is the maximum of finitely many functions of w, namely 0 and the differences $w(us) - w(vs)$ over a set of u that does not depend on w; so both sides are continuous in w. Letting $k \to \infty$ along those k gives (iii) for w and v.

6 Directed-cycle consequences

In an oriented graph with no directed triangle, every exact second out-neighbor is a non-neighbor. Combining this observation with Theorem 1.1 gives degree and order bounds. The first assertion below is the oriented-graph form of Thomassé's non-neighbor statement recorded by Sullivan [21, Conjecture 6.12]. The consequence using both minimum indegree and minimum outdegree is also noted in [21, Section 3]. The regular case gives the girth-four lower bound in the Behzad–Chartrand–Wall minimum-order conjecture [1], also recorded by Sullivan [21, Section 4].

Corollary 6.1 (Non-neighbors and directed cycles). *Let D be a nonempty finite oriented graph of order n. For $v \in V(D)$, write*

$$N_1^-(v) = \{u : u \to v\}, \qquad \deg^{\pm}(v) = |N_1^{\pm}(v)|, \qquad \delta^{\pm}(D) = \min_{x \in V(D)} \deg^{\pm}(x).$$

Define the non-neighbors of v, excluding v itself, by

$$\mathrm{Non}(v) = V(D) \setminus \big(\{v\} \cup N_1^+(v) \cup N_1^-(v)\big).$$

Then the following hold.

(i) *If D has no directed 3-cycle, then some vertex v satisfies*

$$\deg^+(v) \le |\mathrm{Non}(v)|, \qquad n-1 \ge 2\deg^+(v) + \deg^-(v).$$

(ii) *If $\delta^+(D) \ge n/3$ and $\delta^-(D) \ge n/3$, then D contains a directed 3-cycle.*

(iii) *If $\varrho \ge 1$ is an integer, $\deg^+(v) = \deg^-(v) = \varrho$ for every vertex, and D has directed girth exactly four, then $n \ge 3\varrho + 1$.*

Here directed girth is the length of a shortest directed cycle.

Proof. (i). Suppose that D has no directed 3-cycle. Let $w \in N_2^+(v)$ and choose a path $v \to u \to w$. By definition $w \ne v$ and $w \notin N_1^+(v)$. If $w \in N_1^-(v)$, the arc $w \to v$ would close the directed 3-cycle $v \to u \to w \to v$. Thus $N_2^+(v) \subseteq \mathrm{Non}(v)$ for every v. Theorem 1.1 gives a vertex v with

$$\deg^+(v) \le |N_2^+(v)| \le |\mathrm{Non}(v)|.$$

Since D is oriented, $N_1^+(v)$ and $N_1^-(v)$ are disjoint, and neither contains v. Hence $V(D) \setminus \{v\}$ is the disjoint union of $N_1^+(v)$, $N_1^-(v)$ and $\mathrm{Non}(v)$, and

$$n - 1 = \deg^+(v) + \deg^-(v) + |\mathrm{Non}(v)| \ge 2\deg^+(v) + \deg^-(v).$$

(ii). If D had no directed 3-cycle, the vertex from (i) would give

$$n - 1 \ge 2\delta^+(D) + \delta^-(D) \ge 3 \cdot \frac{n}{3} = n,$$

a contradiction. (As degrees are integers, the hypothesis is equivalent to $\delta^{\pm}(D) \ge \lceil n/3 \rceil$; the case of equality when 3 divides n is included.)

(iii). Directed girth four excludes directed 3-cycles, so (i) applies; in the regular case its second inequality reads $n - 1 \ge 3\varrho$. □

7 Context and prior work

The conjecture was recorded by Dean and Latka [8]. A *tournament* is an oriented graph with an arc between every pair of distinct vertices. This case, also known as Dean's conjecture, was proved by Fisher [12] using a probability distribution obtained from Farkas' lemma. Havet and Thomassé gave a combinatorial proof using local median orders [14]. Fidler and Yuster [11] extended the tournament theorem to tournaments missing a matching, a star, or a clique. Ghazal corrected the star-case proof and treated generalized stars [13]. Subsequent advances include a matching together with a star [7] and two stars [6].

Other approaches impose degree conditions or weaken the desired ratio. Kaneko and Locke established the conjecture for minimum outdegree at most six [16]; a recent computer-assisted preprint treats minimum outdegree seven [18]. Chen, Shen and Yuster proved a universal second-to-first neighborhood ratio of approximately 0.657298 [5]. Huang and Peng raise this to approximately 0.715538 and additionally report a computational bound of 0.74530 [15]. Botler, Moura and Naia [2], followed by Espuny Díaz, Girão, Granet and Kronenberg [10], studied every orientation of binomial random graphs; the latter result covers every fixed edge probability below $1/2$, with probability tending to one as the order tends to infinity. Brukhman's recent dense-case preprint concerns graphs of order $n \le 2\delta + 2$, where δ is the minimum outdegree [4].

The proof given here builds on the minimal-counterexample and graph-product methods of Brantner, Brockman, Kay and Snively [3] and on Seacrest's subset perspective [20]. The edge-deletion idea in the proof of Seacrest's Lemma 4 is adapted in Lemma 2.2. The main additional ingredient is the pruning bound of Lemma 3.1 for arbitrary finite binary relations, whose deletion cost depends on the points left uncovered after pruning.

8 Verification checklist

Status. Every step was checked while writing and none failed, so no step carries an "Unverified step" mark. This records a line-by-line reading, not an independent certification; the list below says where to look first.

Proved here versus cited. The proof of Theorem 1.1 (Sections 2–4) is self-contained: the facts attributed to Edmonds and Kőnig are proved in Parts 6 and 10 of the proof of Lemma 3.1, and no result of Seacrest or of Brantner et al. is invoked. Only the corollaries use outside results: Corollary 5.1(iii) rests on [19, Proposition 2] (Remark 5.2 is an optional substitute), and part (ii) uses Farkas' lemma as quoted there.

1. **Bookkeeping in the arc-deletion argument** (Lemma 2.2, Steps (5d)–(5h)): the inclusion (7), the inclusion $\widetilde{N}_2(v) \subseteq (N_2^+(v) \setminus \Psi) \cup (Y \cap F(T))$, and the exact identity $g(T \cup Y) - g(T) = |\Psi| - |Y \cap F(T)|$. Also check that the witness v really loses an out-neighbour (5b), and that the increments telescope over disjoint sets Y_k (Step 6).

2. **The second-image bound in the blow-up** (Lemma 2.3, Step 5) and the arithmetic of Step 6 leading to $m(|F^2(S) \setminus F(S)| - |F(S) \setminus S|) + n$; then the separate case $S = V(D)$ (Step 8).

3. **Vertex cover ⇒ pruning bound** (Lemma 3.1, Claims 1 and 2): conflict-free members are isolated, and each covered point of Z forces a distinct copy into K, giving (14).

4. $NL = AB$ **and the rank chain** (Parts 4 and 5): (16) entry by entry, and (18).

5. **One real assignment for all genericity conditions** (Part 6): (6a) uses Claim 3 (the two sub-matchings are maximum); (6b)–(6c) use the block-by-block non-cancellation; (6d) shows that a maximum-rank minor of $W^{p,i}$ is a nonzero polynomial; (6e) takes a non-root of the product.

6. **The kernel restriction** (Part 8): each row equation (22) is either trivial or a genuine coordinate of $B\mathbf{u} = 0$ because $(p, s') \in H$; likewise for (23) and Z. Then Lemma 3.3 applies to the fixed real matrix $W^{p,i}$. Part 9 turns this into $\dim \ker A \geq k$.

7. **The three facts feeding the pruning lemma in Proposition 4.1**: the diagonal lies in $\mathcal{R} \cap \mathcal{C}$ and is conflict-free (Steps 5–6), $|H| \leq d$ (Step 7), and $q \leq d'$ (Step 8); then the final arithmetic $|P| + |Q| + 2d - (d + d') + d' = |P| + |Q| + d$.

8. **Where orientedness is used**: identity (1); the minimal counterexample in Lemma 2.2; G oriented (Lemma 2.3, Step 2); compatibility of (Δ, Δ) (Proposition 4.1, Step 1). Lemma 3.1 itself uses no orientation hypothesis.

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